

Lab2

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.1
## Warning: package 'ggplot2' was built under R version 4.5.2
## Warning: package 'tidyr' was built under R version 4.5.1
## Warning: package 'readr' was built under R version 4.5.1
## Warning: package 'purrr' was built under R version 4.5.1
## Warning: package 'dplyr' was built under R version 4.5.1
## Warning: package 'stringr' was built under R version 4.5.1
## Warning: package 'forcats' was built under R version 4.5.1
## Warning: package 'lubridate' was built under R version 4.5.1
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr    1.5.1
## v ggplot2    4.0.2      v tibble     3.2.1
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(plotrix)
```

```
## Warning: package 'plotrix' was built under R version 4.5.2
```

```
grass <- read_csv("../data_raw/Site 1.csv")
```

```
## Rows: 10 Columns: 3
## -- Column specification -----
## Delimiter: ","
## dbl (3): plant_number, height_cm, weight_g
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

For the height and weight of plants on each site, calculate the mean, variance, standard deviation, coefficient of variation, and the standard error of the mean.

For weight on Site 1:

```

mean(grass$weight_g)

## [1] 9.55
var(grass$weight_g)

## [1] 6.291667
sd(grass$weight_g)

## [1] 2.508319
Mean weight: 9.55 Variance in weight: 6.29 Standard deviation in weight: 2.5 Coefficient of variation:
sd/mean
sd(grass$weight_g)/mean(grass$weight_g)

## [1] 0.2626513
Standard error of the mean
std.error(grass$weight_g)

## [1] 0.7932003
For height on Site 1:
mean(grass$height_cm)

## [1] 24
var(grass$height_cm)

## [1] 43.33333
sd(grass$height_cm)

## [1] 6.582806
Mean height: 24 Variance in height: 43.33 Standard deviation in height: 6.58 Coefficient of variation: sd/mean
sd(grass$height_cm)/mean(grass$height_cm)

## [1] 0.2742836
Standard error of the mean
std.error(grass$height_cm)

## [1] 2.081666
For weight on Site 2:
grass2 <- read_csv("../data_raw/Site 2.csv")

## Rows: 10 Columns: 3
## -- Column specification -----
## Delimiter: ","
## dbf (3): plant_number, height_cm, weight_g
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.

```

```

mean(grass2$weight_g)

## [1] 23.15
var(grass2$weight_g)

## [1] 17.01833
sd(grass2$weight_g)

## [1] 4.125328
Mean weight: 23.15 Variance in weight: 17.02 Standard deviation in weight: 4.13 Coefficient of variation:
sd/mean
sd(grass2$weight_g)/mean(grass2$weight_g)

## [1] 0.1781999
std.error(grass2$weight_g)

## [1] 1.304543
Height for Site 2:
mean(grass2$height_cm)

## [1] 56.1
var(grass2$height_cm)

## [1] 578.5444
sd(grass2$height_cm)

## [1] 24.05295
Mean height: 56.1 Variance in height: 578.54 Standard deviation in height: 24.05 Coefficient of variation:
sd/mean
sd(grass2$height_cm)/mean(grass2$height_cm)

## [1] 0.4287514
std.error(grass2$height_cm)

## [1] 7.606211

```

Using the weight data, determine if the difference between the sites is significant. Give a probability based on a T-test and use a graph to display the difference or lack of difference.

```

t.test(grass$weight_g, grass2$weight_g)

##
## Welch Two Sample t-test
##
## data: grass$weight_g and grass2$weight_g
## t = -8.9077, df = 14.854, p-value = 2.416e-07
## alternative hypothesis: true difference in means is not equal to 0

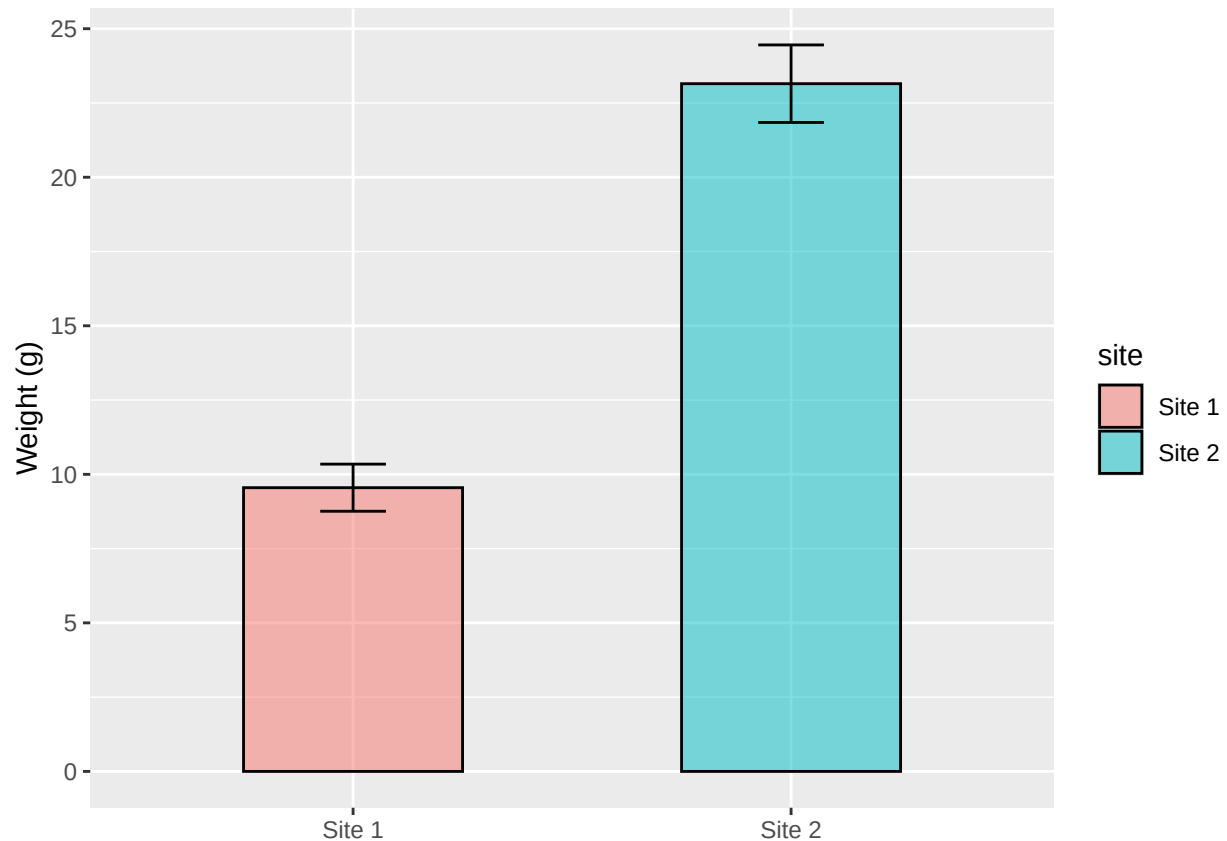
```

```
## 95 percent confidence interval:  
## -16.85699 -10.34301  
## sample estimates:  
## mean of x mean of y  
##      9.55      23.15
```

The means are significantly different (p-value: <0.05).

Graph of the means:

```
site1 <- read.csv("../data_raw/Site 1.csv")  
site2 <- read.csv("../data_raw/Site 2.csv")  
  
#Combine dfs  
site1$site <- "Site 1"  
site2$site <- "Site 2"  
combined_data <- rbind(site1, site2)  
  
# Calculate mean and standard error of the mean (SEM) for weight  
weight_summary <- combined_data %>%  
  group_by(site) %>%  
  summarise(  
    mean_weight = mean(weight_g, na.rm = TRUE),  
    sem_weight = sd(weight_g, na.rm = TRUE) / sqrt(n())  
  )  
  
# Plot mean weight with error bars  
ggplot(weight_summary, aes(x = site, y = mean_weight, fill = site)) +  
  geom_bar(stat = "identity", width = 0.5, color = "black", alpha = 0.5) +  
  geom_errorbar(  
    aes(ymin = mean_weight - sem_weight, ymax = mean_weight + sem_weight),  
    width = 0.15) +  
  labs(x = NULL, y = "Weight (g)")
```



#Graph the relationship between height and weight at each site. Display R^2 value.

--- Site 1 Graph ---

#fit linear model

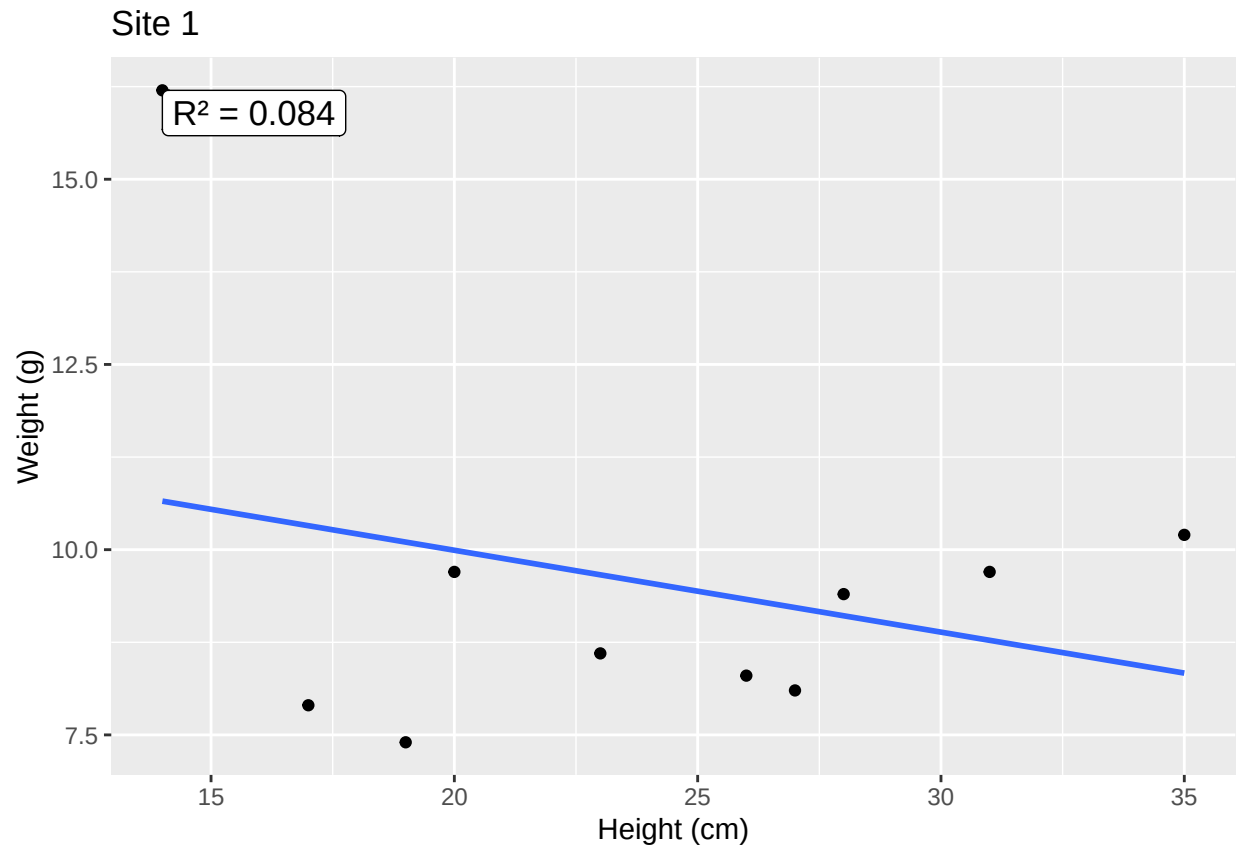
```
r2_site1 <- summary(lm(weight_g ~ height_cm, data = site1))$r.squared
```

#plot scatterplot

```
plot1 <- ggplot(site1, aes(x = height_cm, y = weight_g)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  annotate("label",
    x = min(site1$height_cm),
    y = max(site1$weight_g),
    label = paste0("R² = ", round(r2_site1, 3)),
    hjust = 0, vjust = 1, size = 4.5) +
  labs(title = "Site 1", x = "Height (cm)", y = "Weight (g)")
```

```
print(plot1)
```

`geom\_smooth()` using formula = 'y ~ x'

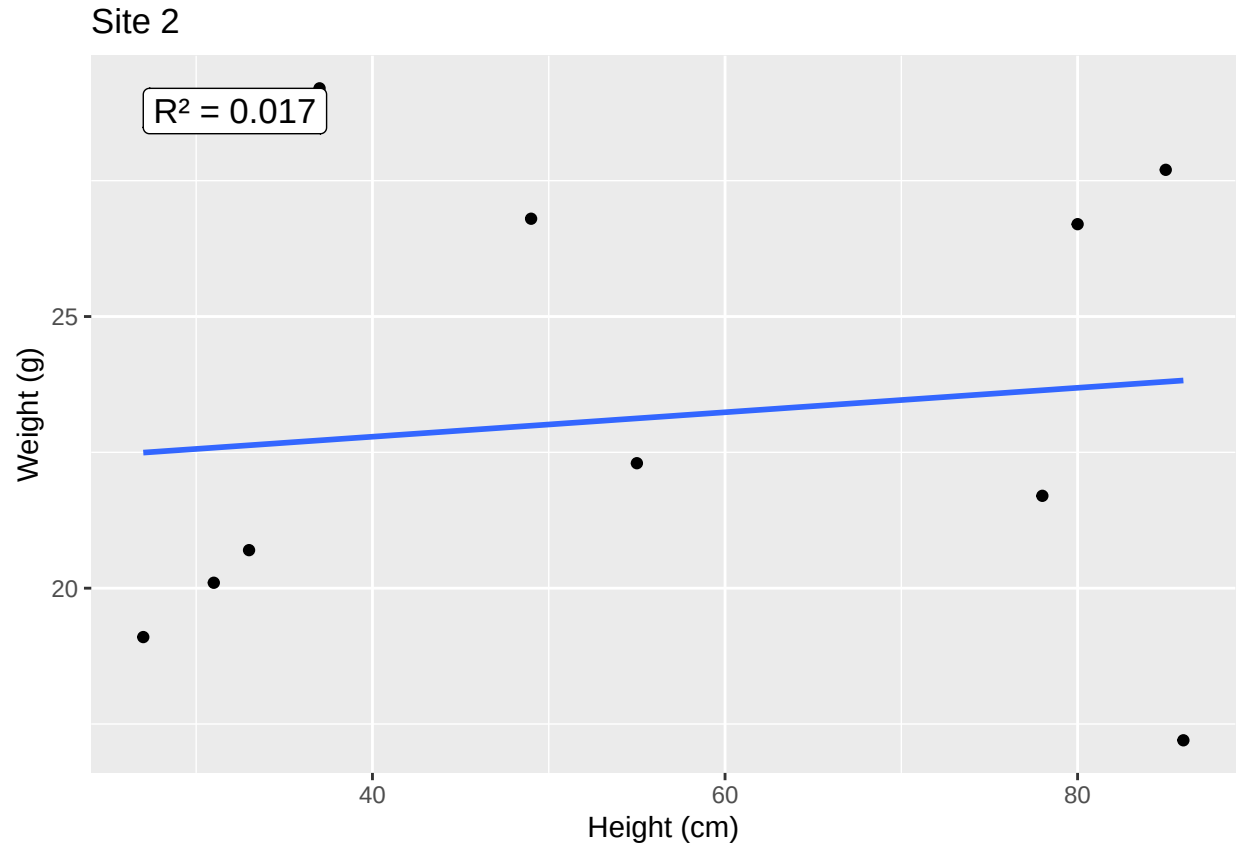


```
# --- Site 2 Graph ---
#fit linear model
r2_site2 <- summary(lm(weight_g ~ height_cm, data = site2))$r.squared

#plot scatterplot
plot2 <- ggplot(site2, aes(x = height_cm, y = weight_g)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  annotate(
    "label",
    x = min(site2$height_cm),
    y = max(site2$weight_g),
    label = paste0("R² = ", round(r2_site2, 3)),
    hjust = 0, vjust = 1, size = 4.5
  ) +
  labs(title = "Site 2", x = "Height (cm)", y = "Weight (g)")

print(plot2)

## `geom_smooth()` using formula = 'y ~ x'
```



#Why would this relationship be important in an inventory and monitoring class?

This is important because relationships like this are predictive. We can infer based on this graph what the height would be if we can measure the weight of plants and vice versa in the future in these sites. However the relationships on these graphs are weakly correlated and would generally be inaccurate if used in this way.

#What are some possible explanations for the similarities and/or differences between the sites? Which seems most likely?

The sites may have different types of plants, different precipitation patterns, occur at different latitudes, have different soil types, face different grazing or herbivory pressures, or different levels of light. For Site 1, if weight decreases as height increases, I imagine there are small shrubs or shortgrasses with tall culms that grow lower to the ground and perhaps lose some herbaceous biomass during their reproductive stage. For Site 2, height and weight are trending together so I imagine these are denser grasses or plants that increase in biomass as they grow taller.

#Calculate the sample size to produce, with 95% probability, a sample mean within 10% of the true population mean for the weights.

$n = ((z*s)/B)^2$ $z = 1.96$ s = Sample standard deviation B = Desired precision level, otherwise known as margin of error

For Site 1:

$n = 27$

For Site 2:

$n = 13$

#Why do we care about sample size? What use is it?

Sample size needs to be large enough to capture the population accurately. Your standard error decreases as sample size increases. Management decisions are based on estimates from samples, and being confident in our sampling is critical in making reliable decisions. If the sample size is too small, standard error will be higher, which means less precision.

#If this were an experiment, how could you properly analyze both height and weight as response variables?

I would use an ANOVA and test weight as a function of height, controlling for site. We want to know if weight affects height, despite site differences.